Predicting Coral Reef Regimes from Human and Natural Influences

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SITES	PREDICTORS	TRAIN	HELD-OUT TEST
620	20	496	124
Hawaiian reef observations	Human + biophysical	80% stratified split	Used once after selection

RESEARCH QUESTION

Can 20 anthropogenic and biophysical predictors classify a Hawaiian reef site into its observed regime label? This is supervised four-class classification using nominal labels **1, 2, 3 and 5**. It predicts the provided label; it does not establish that any predictor causes a regime transition.

DATASET & TARGET

Source: Hawaii_RegimesPredictors.txt
Rows: 620; **inputs:** 20
Predictors: Effluent through Depth
Class counts: 1: 171; 2: 172; 3: 143; 5: 134
Exact duplicate rows: 0; none removed automatically.

LEAKAGE SAFEGUARDS

Excluded: Regime1/2/3/5 target one-hot fields, ecological response variables, IDs and coordinates.

Missingness: Complexity 113 values; Depth 12 values.
Imputation: median imputation inside each pipeline / training fold.
Split: random_state=42; 496 train / 124 test.

EVALUATION DESIGN

01	02	03	04	05
Audit	**Split**	**Compare**	**Select**	**Test**
Validate schema	Stratified 80/20	5-fold training CV	Mean macro-F1	Held-out evaluation

MODELS COMPARED

Most-frequent baseline; class-weighted Logistic Regression; RBF SVM; Random Forest; Gradient Boosting; and Random Forest plus five interaction features. The selected model is chosen only from training-set cross-validation; the held-out test set is reserved for final evaluation.

Evaluation choice: macro-F1 gives equal weight to each class F1, making it useful when class-level performance matters. Accuracy, balanced accuracy, micro-F1, weighted-F1 and per-class results are also reported.

Scope note: this is an adapted, reproducible implementation of the assigned problem, not an exact reproduction of every experiment in the Stanford reference project.

RESULTS & INTERPRETATION

Random Forest led cross-validation and held-out evaluation

BEST CV MACRO-F1
0.692
Random Forest; mean across 5 folds

TEST MACRO-F1
0.702
n = 124; held-out split

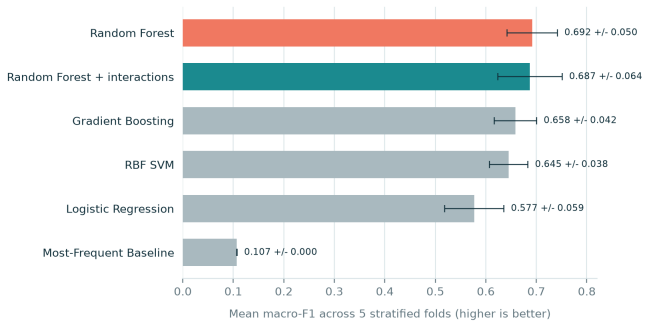
TEST ACCURACY
0.702
87 of 124 correct

TRAINING-ONLY CROSS-VALIDATION COMPARISON

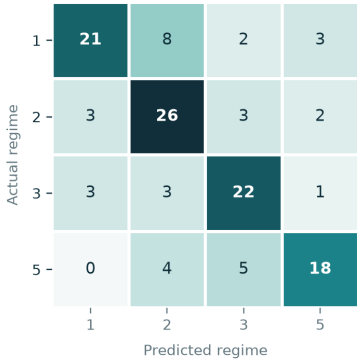
Model	Input features	Macro-F1 mean	Fold SD	Balanced acc.	Accuracy
Random Forest	20	0.692	0.050	0.692	0.698
Random Forest + interactions	25	0.687	0.064	0.686	0.694
Gradient Boosting	20	0.658	0.042	0.660	0.667
RBF SVM	20	0.645	0.038	0.647	0.649
Logistic Regression	20	0.577	0.059	0.577	0.585
Most-Frequent Baseline	-	0.107	0.000	0.250	0.272

Model selection uses only training-set CV macro-F1. The interaction ablation (0.687) did not outperform the base Random Forest (0.692).

CROSS-VALIDATION



HELD-OUT TEST



CLASS-WISE RECALL

Regime 1 Recall 0.618 | F1 0.689

Regime 2 Recall 0.765 | F1 0.693

Regime 3 Recall 0.759 | F1 0.721

Regime 5 Recall 0.667 | F1 0.706

TOP PERMUTATION-IMPORTANCE FEATURES

1. PAR_STD
Mean macro-F1 decrease when shuffled: 0.023

2. CHL_CLIM_M
Mean macro-F1 decrease when shuffled: 0.018

3. Fishing_NonComm_Boat_Total
Mean macro-F1 decrease when shuffled: 0.015

Limitations: the sample is modest and restricted to Hawaii; regime 4 is absent in the provided subset; the model has not been validated on independent geographic data. Permutation importance is descriptive, not causal.

References: Jouffray et al. (2019), *Parsing human and biophysical drivers of coral reef regimes*, Proc. R. Soc. B 286:20182544, https://doi.org/10.1098/rspb.2018.2544; Stanford CS229 project: https://cs229.stanford.edu/proj2019spr/report/56.pdf; data: https://github.com/JBjouffray/Hawaii_RegimesPredictors.